

The GenAI Personal Lab ? sample document

This is a short sample document used to exercise the RAG demos in this project. It has five pages, each covering one retrieval technique in plain language, so a question about a specific technique should retrieve the matching page.

Naive RAG

Naive RAG is the simplest retrieval pipeline: split a document into chunks, embed each chunk, store the embeddings, and at question time embed the question and run a vector search for the closest chunks. Those chunks are placed in a prompt and a language model writes the answer. It has no re-ranking, no query rewriting, and no agentic control -- it is a straight line from question to answer.

Chunking strategies

How a document is split changes what gets retrieved. Fixed-size chunking cuts text every N characters. Recursive chunking tries to split on paragraph and sentence boundaries first. Sliding-window chunking overlaps neighbouring chunks so a fact near a boundary is not lost. Semantic chunking splits where the meaning changes, detected by a drop in embedding similarity between neighbouring sentences.

Hybrid search

Hybrid search runs a vector search and a keyword (BM25) search separately, then fuses the two ranked lists, commonly with reciprocal rank fusion. This helps with queries that name an exact code, identifier, or proper noun, which vector search alone can miss because embeddings capture meaning rather than exact tokens.

RAG evaluation

Evaluating a RAG pipeline means scoring both retrieval and generation. Faithfulness checks whether the answer is supported by the retrieved context. Answer relevancy checks whether the answer addresses the question. Context precision and context recall check whether the retrieved passages are both correct and sufficient.